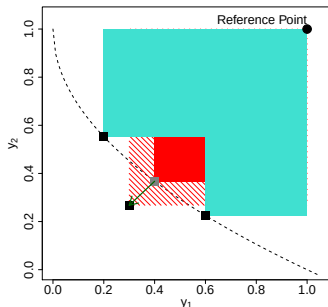
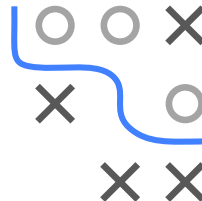


Optimization in Machine Learning

Bayesian Optimization

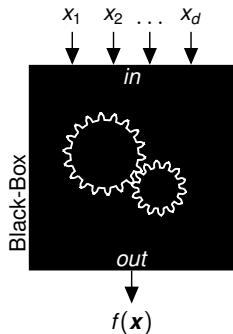
Multicriteria Bayesian Optimization



Learning goals

- Multicriteria Optimization
- Taxonomy
- ParEGO, SMS-EGO, EHI

MULTICRITERIA BAYESIAN OPTIMIZATION



$$f : \mathcal{S} \rightarrow \mathbb{R}^m$$

$$\min_{\mathbf{x} \in \mathcal{S}} f(\mathbf{x}) = (f_1(\mathbf{x}), \dots, f_m(\mathbf{x}))$$

- A configuration $\mathbf{x}$ **dominates** ($\prec$) $\tilde{\mathbf{x}}$ if

$$\forall i \in \{1, \dots, m\} : f_i(\mathbf{x}) \leq f_i(\tilde{\mathbf{x}})$$

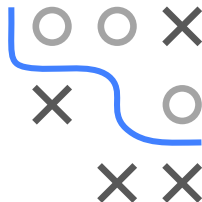
and $\exists j \in \{1, \dots, m\} : f_j(\mathbf{x}) < f_j(\tilde{\mathbf{x}})$

- Set of non-dominated solutions:

$$\mathcal{P} := \{\mathbf{x} \in \mathcal{S} \mid \nexists \tilde{\mathbf{x}} \in \mathcal{S} : \tilde{\mathbf{x}} \prec \mathbf{x}\}$$

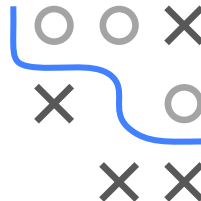
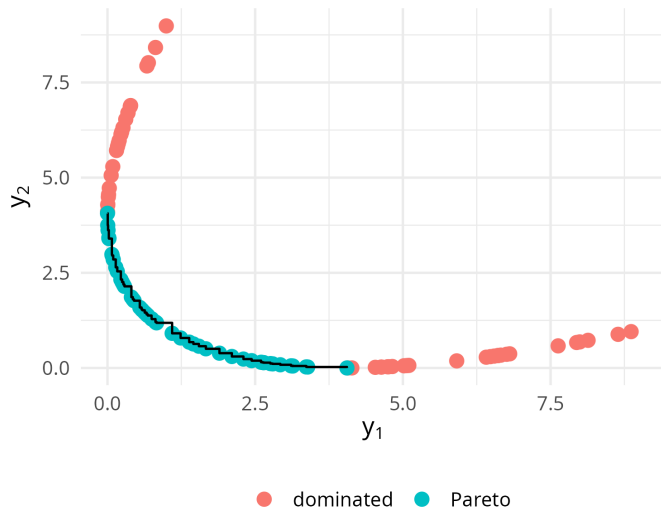
- Pareto set $\mathcal{P}$, Pareto front $\mathcal{F} = f(\mathcal{P})$

- Goal: Find $\hat{\mathcal{P}}$ of non-dominated points that estimates the true Pareto set $\mathcal{P}$



MULTICRITERIA BAYESIAN OPTIMIZATION

Example Pareto front:



MULTICRITERIA BAYESIAN OPTIMIZATION

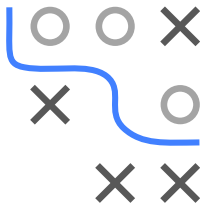
The most popular quality indicator is the hypervolume indicator (also called dominated hypervolume or $\mathcal{S}$ -metric).

The hypervolume, HV, of an approximation of the Pareto front $\hat{\mathcal{F}} = f(\hat{\mathcal{P}})$ can be defined as the combined volume of the dominated hypercubes domHC_r of all solution points $\mathbf{x} \in \hat{\mathcal{P}}$ regarding a reference point $\mathbf{r}$, i.e.,

$$\text{HV}_r(\hat{\mathcal{P}}) := \mu \left(\bigcup_{\mathbf{x} \in \hat{\mathcal{P}}} \text{domHC}_r(\mathbf{x}) \right)$$

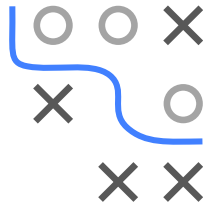
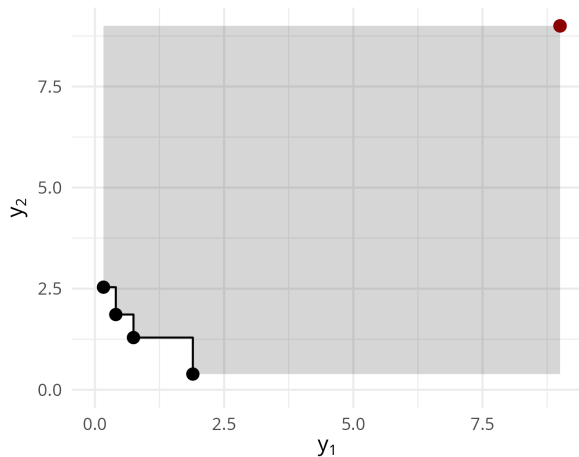
where μ is the Lebesgue measure and the dominated hypercube is given as:

$$\text{domHC}_r(\mathbf{x}) := \{ \mathbf{u} \in \mathbb{R}^m \mid f_i(\mathbf{x}) \leq u_i \leq r_i \forall i \in \{1, \dots, m\} \}$$



MULTICRITERIA BAYESIAN OPTIMIZATION

Hypervolume example:



Reference point $\mathbf{r}$ in red, estimated Pareto front $\hat{\mathcal{F}}$ in black,
corresponding $\text{HV}_{\mathbf{r}}(\hat{\mathcal{P}})$ is given by the grey area

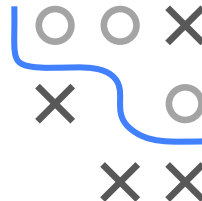
```
graph TD
    subgraph Initial_design [Initial design]
        A1[Arbitrary DOE  
(e. g. random,  
Sobol, grid)]
        A2[Latin Hypercube Sampling]
    end

    subgraph Model_fitting [Model fitting]
        B1[Individual models for each objective]
        B2[Scalarize objectives]
        B3[Model of scalarization]
        A1 --> B1
        A1 --> B2
        A2 --> B3
        B2 --> B3
    end

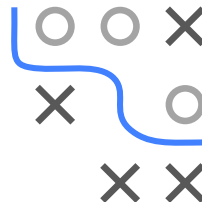
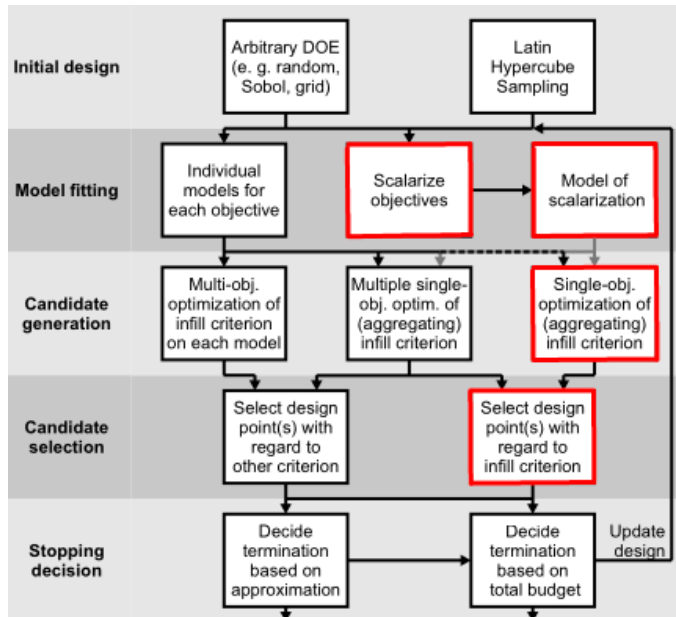
    subgraph Candidate_generation [Candidate generation]
        C1[Multi-obj. optimization of infill criterion on each model]
        C2[Multiple single-obj. optim. of (aggregating) infill criterion]
        C3[Single-obj. optimization of (aggregating) infill criterion]
        B1 --> C1
        B2 --> C2
        B3 --> C2
        B3 --> C3
    end

    subgraph Candidate_selection [Candidate selection]
        D1[Select design point(s) with regard to other criterion]
        D2[Select design point(s) with regard to infill criterion]
        C1 --> D1
        C2 --> D1
        C2 --> D2
        C3 --> D2
    end

    subgraph Stopping_decision [Stopping decision]
        E1[Decide termination based on approximation]
        E2[Decide termination based on total budget]
        E3[Update design]
        D1 --> E1
        D2 --> E1
        D2 --> E2
        E1 --> E2
        E2 --> E3
        E3 --> A2
    end
```



PAREGO



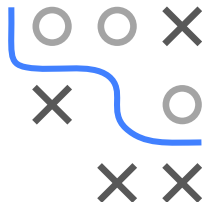
PAREGO

- 1 Scalarize standardized objectives using the augmented Tchebycheff norm

$$\max_{i \in \{1, \dots, m\}} w_i f_i(\mathbf{x}) + \rho \sum_{i=1}^m w_i f_i(\mathbf{x})$$

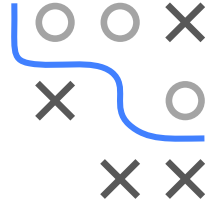
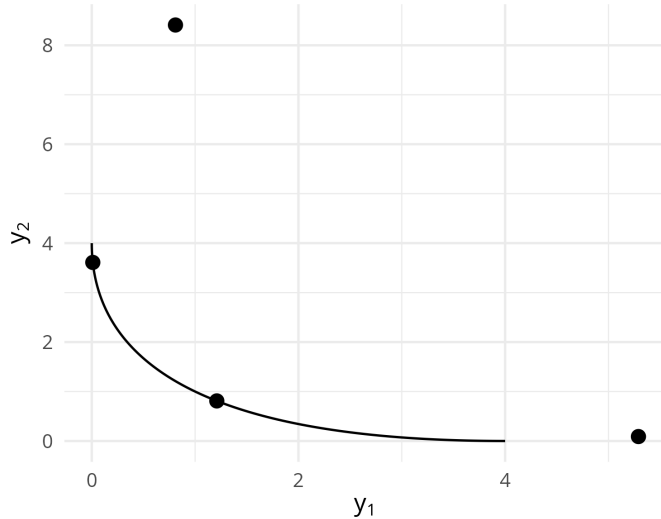
with weight vector $\mathbf{w}$ drawn uniformly from the set of evenly distributed weight vectors $\mathcal{W}$

- 2 Fit SM on the scalarized objective function
- 3 Proceed to use any standard single-objective acquisition function (EI, PI, LCB, ...)



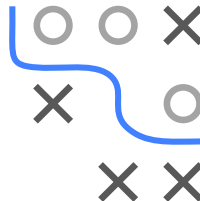
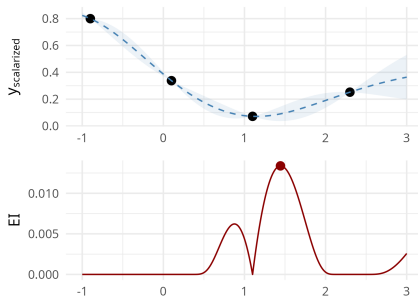
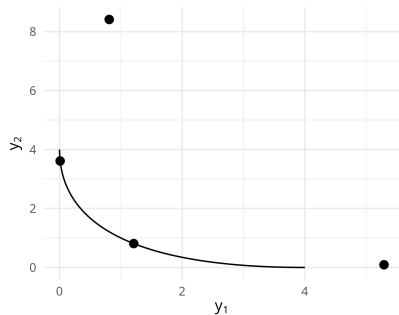
PAREGO

ParEGO Example, initial design and true Pareto front in black ...



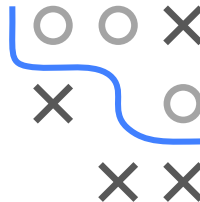
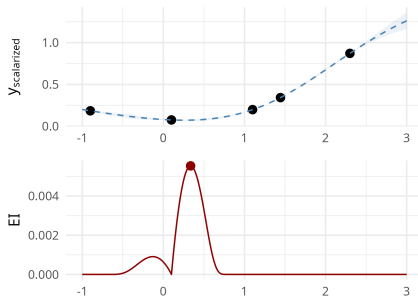
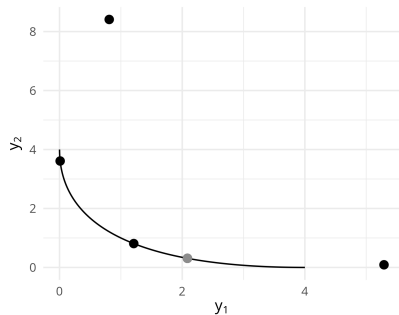
PAREGO

... standardize objectives, obtain scalarized objective via augmented Tchebycheff norm, fit SM and optimize EI ...



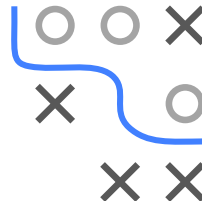
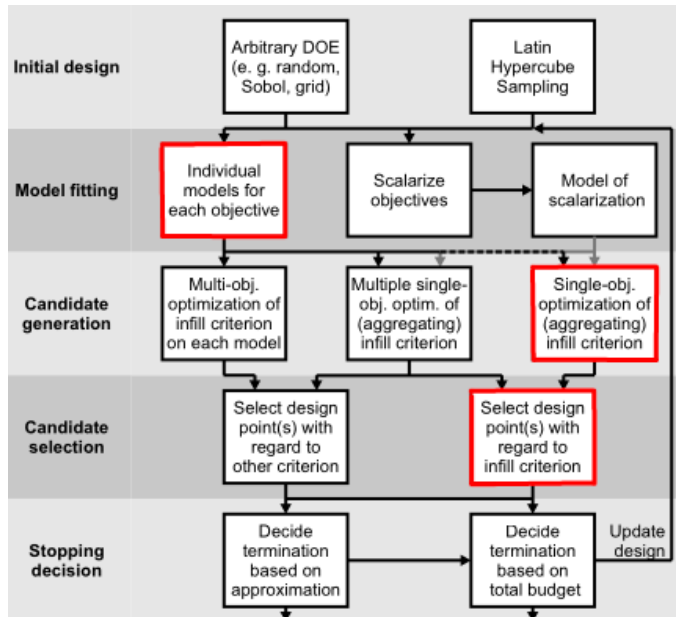
PAREGO

... note that the specific scalarization is different at each iteration!



The grey point visualizes the candidate we choose to evaluate in the previous iteration

SMS-EGO

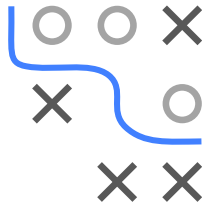
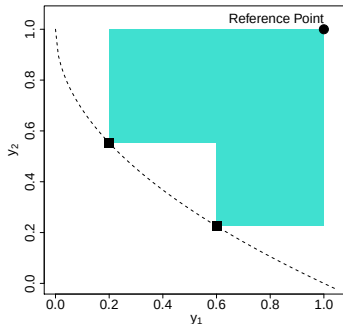


SMS-EGO

Individual models for each objective f_i

Single-objective optimization of aggregating acquisition function:
Calculate contribution of the confidence bound of candidate to the current front approximation

- Calculate LCB for each objective
- Measure contribution with regard to the hypervolume improvement
- For ε -dominated ($\prec_\varepsilon$) solutions, a penalty is added

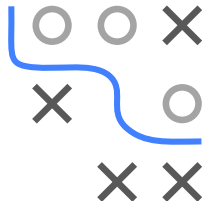
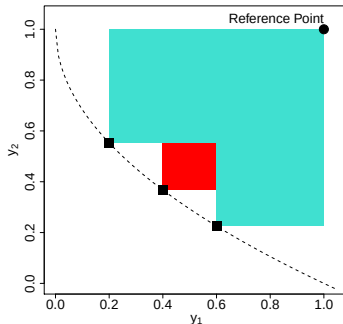


SMS-EGO

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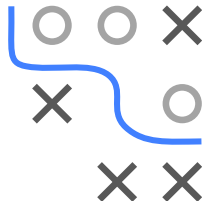
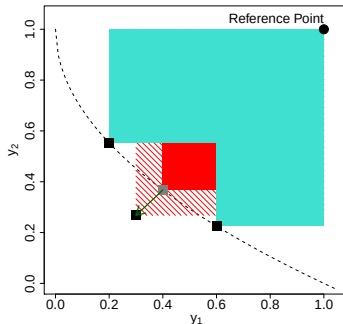


SMS-EGO

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OUTLOOK

- Many more options exist:
 - Expected Hypervolume Improvement
 - Multi-EGO
 - Entropy based: PESMO, MESMO
 - ...

